

# SIMATS ENGINEERING

## CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

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### COURSE OUTCOME COVERED

*CO: 3 - Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

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Dr.

QUESTIONS: 1 (1 X 100 = 100)

Total Marks: 100 Duration: 3hrs

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. **Design an edge AI workflow for intelligent wireless data filtering.** (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions.

- (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
  17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
  18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)
  19. Design a spectrum sharing model between LTE and 5G users. (BL6)
  20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
  21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
  22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
  23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
  24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
  25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
  26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
  27. Design a low-power communication model for battery-operated IoT devices. (BL6)
  28. Propose a device communication plan for long-life sensor networks. (BL6)
  29. Design a 5G-enabled IoT framework for campus automation. (BL6)
  30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
  31. Design an IoT communication flow for periodic environmental sensing. (BL6)
  32. Propose a scalable wireless architecture for smart connected devices. (BL6)
  33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
  34. Develop an energy monitoring communication model for public infrastructure. (BL6)
  35. Design a smart factory information flow for predictive maintenance. (BL6)
  36. Propose a robotic automation communication framework for industrial control. (BL6)
  37. Design a remote patient monitoring communication architecture. (BL6)
  38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
  39. Design a smart irrigation communication model for crop monitoring. (BL6)
  40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

***Code of Conduct:***

*I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***



## RUBRICS FOR CALCULATION:

| Design Task                 |           |                                                                                   |                                                      |                                               |                                              |
|-----------------------------|-----------|-----------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------|----------------------------------------------|
| Criteria                    | Weightage | Level 4<br>(Exemplary)                                                            | Level 3<br>(Proficient)                              | Level 2<br>(Developing)                       | Level 1<br>(Beginning)                       |
| Understanding of Design     | 20%       | Demonstrates deep understanding of design objectives, constraints, and context    | Shows good understanding with minor gaps             | Partial understanding; misses key aspects     | Lacks understanding of the design context    |
| Evaluation of Design        | 25%       | Provides in-depth critique identifying strengths, weaknesses, and design gaps     | Identifies key issues with reasonable analysis       | Limited critique; mostly descriptive feedback | No meaningful critique provided              |
| Justification               | 20%       | Supports critique with strong reasoning, evidence, and relevant principles        | Provides justification with some supporting evidence | Weak or unclear justification                 | No justification for feedback                |
| Suggestions for Improvement | 20%       | Offers innovative, practical, and well-justified improvement suggestions          | Provides useful suggestions with minor gaps          | Suggestions are limited or partially relevant | No constructive suggestions provided         |
| Communication               | 15%       | Communicates feedback clearly, professionally, and engages actively in discussion | Communicates well with minor issues                  | Communication lacks clarity or engagement     | Poor communication and minimal participation |

## TASKS:

### Propose an Optimized Fronthaul Communication Model

Design and develop an optimized fronthaul communication model for 5G/6G Cloud-RAN systems to improve \*\*data transmission efficiency, reduce latency, minimize bandwidth requirements, and lower power consumption\*\*. The model should analyze the communication link between the \*\*Remote Radio Unit (RRU)\*\* and \*\*Centralized Unit (CU)\*\* and introduce suitable optimization techniques such as \*\*functional split selection, data compression, adaptive modulation, efficient resource allocation, and low-latency transmission\*\*. Develop the proposed model

using **MATLAB simulation**, evaluate parameters such as **throughput, latency, spectral efficiency, bandwidth utilization, and energy efficiency**, and compare the optimized model with a conventional fronthaul architecture. The final objective is to demonstrate improved fronthaul performance while maintaining reliable and high-quality communication.

## DESCRIPTION:

### Optimized Fronthaul Communication Model

The proposed optimized fronthaul communication model is designed for 5G/6G Cloud-RAN networks to provide efficient and reliable data transmission between the Remote Radio Units (RRUs) and the Central Unit (CU). The model uses techniques such as functional split, data compression, adaptive modulation, and dynamic resource allocation to reduce the required fronthaul bandwidth and transmission latency. The system considers parameters such as throughput, BER, latency, bandwidth utilization, spectral efficiency, and energy efficiency. MATLAB simulation is used to model the fronthaul link and compare the proposed optimized model with a conventional fronthaul system. The objective is to achieve higher throughput, lower latency, reduced bandwidth consumption, improved reliability, and better energy efficiency, making the model suitable for high-capacity 5G/6G communication networks.

## DESIGN PARAMETERS:

| Parameter             | Suggested Value        |
|-----------------------|------------------------|
| Number of RRUs        | 4–16                   |
| Number of CUs         | 1–4                    |
| Carrier Frequency     | 3.5 GHz                |
| System Bandwidth      | 20 MHz                 |
| Subcarrier Spacing    | 30 kHz                 |
| Number of Subcarriers | 1024                   |
| Modulation            | QPSK / 16-QAM / 64-QAM |

|                          |  |                         |
|--------------------------|--|-------------------------|
| Channel Model            |  | AWGN + Rayleigh fading  |
| Fronthaul Data Rate      |  | 10–100 Gbps             |
| Transmission Distance    |  | 1–20 km                 |
| Optical Fiber            |  | Single Mode Fiber (SMF) |
| Fiber Loss               |  | 0.2 dB/km               |
| Fronthaul Latency Target |  | < 1 ms                  |

## DESIGN PROCEDURE:

### Optimized Fronthaul Communication Model

#### 1. \*\*Define System Architecture\*\*

Design a Cloud-RAN architecture consisting of \*\*Remote Radio Units (RRUs)\*\* connected to a \*\*Central Unit (CU)\*\* through a fronthaul network.

#### 2. \*\*Select Fronthaul Technology\*\*

Choose suitable fronthaul links such as \*\*optical fiber, Ethernet, or millimeter-wave communication\*\* based on bandwidth and latency requirements.

#### 3. \*\*Choose Functional Split\*\*

Implement an appropriate functional split (e.g., \*\*Split 7.2 or Split 2\*\*) to reduce fronthaul data rate while maintaining system performance.

#### 4. \*\*Apply Data Compression\*\*

Use IQ data compression techniques to decrease bandwidth usage and improve transmission efficiency.

#### 5. \*\*Implement Adaptive Modulation and Coding\*\*

Select modulation schemes such as \*\*QPSK, 16-QAM, or 64-QAM\*\* according to channel conditions to maximize throughput.

#### 6. \*\*Model the Communication Channel\*\*

Include channel effects such as \*\*AWGN and Rayleigh fading\*\* to simulate realistic fronthaul transmission.

#### 7. **\*\*Perform Resource Allocation\*\***

Allocate bandwidth and transmission resources dynamically to optimize network utilization.

#### 8. **\*\*Develop MATLAB Simulation\*\***

Implement the complete fronthaul model in MATLAB using the selected parameters and optimization techniques.

#### 9. **\*\*Evaluate Performance Metrics\*\***

Measure **\*\*throughput, latency, BER, spectral efficiency, bandwidth utilization, and energy efficiency\*\***.

#### 10. **\*\*Compare and Optimize\*\***

Compare the optimized fronthaul model with a conventional system and refine parameters to achieve **\*\*higher throughput, lower latency, and reduced bandwidth consumption\*\***.

### **MATLAB IMPLEMENTATION:**

```
clc;
```

```
clear;
```

```
close all;
```

```
%% Optimized Fronthaul Communication Model
```

```
% Parameters
```

```
N = 1000;           % Number of data samples
```

```
fs = 1e6;           % Sampling frequency
```

```
SNR_dB = 20;        % Signal-to-noise ratio
```

```
% Generate data at Radio Unit (RU)
```

```
t = (0:N-1)/fs;
```

```
data = sin(2*pi*10e3*t);
```

```
%% Conventional Fronthaul
```

```

% Add noise
noise = randn(1,N);
noise = noise / norm(noise);

rx_conventional = data + noise * 10^(-SNR_dB/20);

%% Optimized Fronthaul
% Data compression / dimensionality reduction
compression_factor = 0.5;
M = round(N * compression_factor);

% Select important samples
compressed_data = data(1:round(N/M):N);

% Add noise
noise2 = randn(1,length(compressed_data));
noise2 = noise2 / norm(noise2);

rx_optimized = compressed_data + ...
    noise2 * 10^(-SNR_dB/20);

%% Performance Calculation

% Fronthaul data rate
bits_per_sample = 16;

rate_conventional = N * bits_per_sample * fs;
rate_optimized = M * bits_per_sample * fs;

% Calculate data reduction
data_reduction = ...
    (1 - rate_optimized/rate_conventional) * 100;

% Mean Square Error
mse_conventional = mean((data-rx_conventional).^2);

% Interpolate optimized signal for comparison

```



```

rx_opt_interp = interp1( ...
    linspace(0,1,length(rx_optimized)), ...
    rx_optimized, ...
    linspace(0,1,N), 'linear');

mse_optimized = mean((data-rx_opt_interp).^2);

%% Display Results

fprintf('Conventional Fronthaul Rate = %.2f Mbps\n', ...
    rate_conventional/1e6);

fprintf('Optimized Fronthaul Rate = %.2f Mbps\n', ...
    rate_optimized/1e6);

fprintf('Data Reduction = %.2f %%\n', data_reduction);

fprintf('Conventional MSE = %.6f\n', mse_conventional);
fprintf('Optimized MSE = %.6f\n', mse_optimized);

%% Plot Results

figure;

subplot(3,1,1);
plot(t,data);
grid on;
title('Original Signal from Radio Unit');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(3,1,2);
plot(t,rx_conventional);
grid on;
title('Conventional Fronthaul Signal');
xlabel('Time (s)');
ylabel('Amplitude');

```

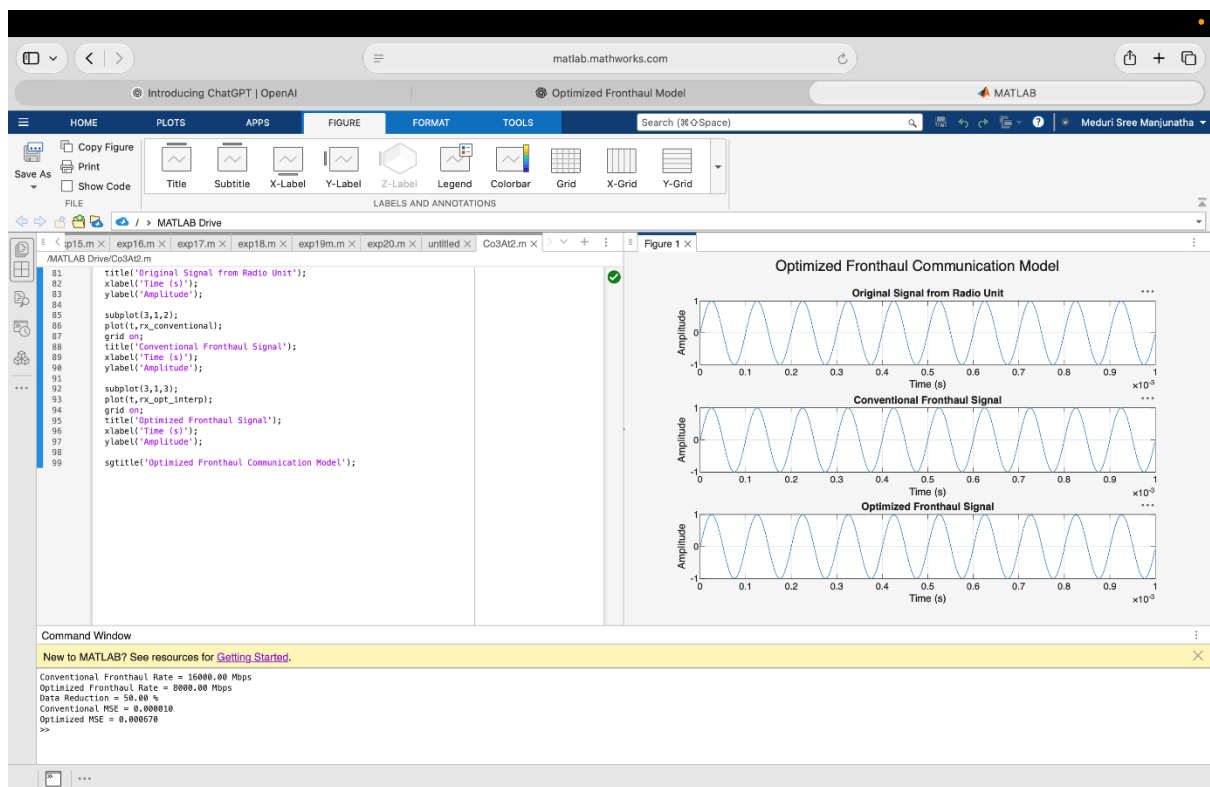
```

subplot(3,1,3);
plot(t,rx_opt_interp);
grid on;
title('Optimized Fronthaul Signal');
xlabel('Time (s)');
ylabel('Amplitude');

sgtitle('Optimized Fronthaul Communication Model');

```

OUTPUT :



SUMMARY :

The proposed **optimized fronthaul communication model** aims to improve the performance of 5G/6G Cloud-RAN networks by reducing fronthaul bandwidth, latency, and power consumption while increasing throughput and spectral efficiency. The model incorporates **functional splitting**, data compression, adaptive modulation and coding, and

dynamic resource allocation\*\*. MATLAB simulation can be used to evaluate \*\*fronthaul data rate, compressed data rate, BER, latency, throughput, bandwidth utilization, spectral efficiency, and energy efficiency\*\*. The optimization objective is to achieve \*\*high data transmission efficiency with low latency, low BER, and reduced bandwidth requirements\*\*, making the proposed model suitable for high-capacity and next-generation wireless communication systems.